AreaClock

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A `Clock` that affects every `Timeline` within its collider by multiplying its time scale with that of their observed clock.

Area clocks can be moved, scaled and rotated at runtime. They can even stack and combine their effects!

Properties

```
ClockBlend innerBlend { get; set; } = Multiplicative
```

Determines how the clock combines its time scale with that of the timelines within.

Value	Description
Multiplicative	The area clock's time scale is multiplied with that of the timelines.
Additive	The area clock's time scale is added to that of the timelines.

In most cases, multiplicative blend will yield the expected results. However, additive blend becomes extremely useful to affect your timelines when they have a time scale of 0.

```
AreaClockMode mode { get; set; }
```

Determines how objects should behave when progressing within the area clock.

Value	Description

Value	Description
Instant	Objects that enter the clock are instantly affected by its full time scale, without any smoothing.
PointToEdge	Objects that enter the clock are progressively affected by its time scale, depending on a A / B ratio where: • A is the distance between `center` and the object • B is the distance between `center` and the collider's edge in the object's direction
DistanceFromEntry	Objects that enter the clock are progressively affected by its time scale, depending on a A / B where: • A is the distance between the object's entry point and its current position • B is the value of `padding`

See the Notes section below for diagrams and more explanations for these modes.

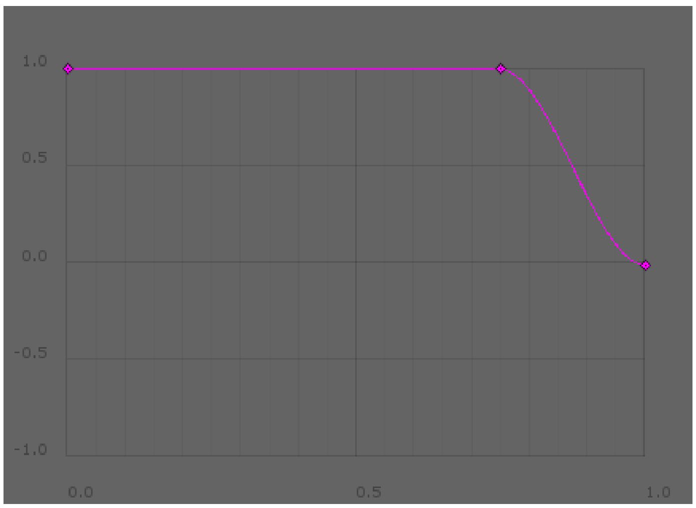
```
AnimationCurve curve { get; set; }
```

The curve of the area clock. This value is only used for the `PointToEdge` and `DistanceFromEntry` modes.

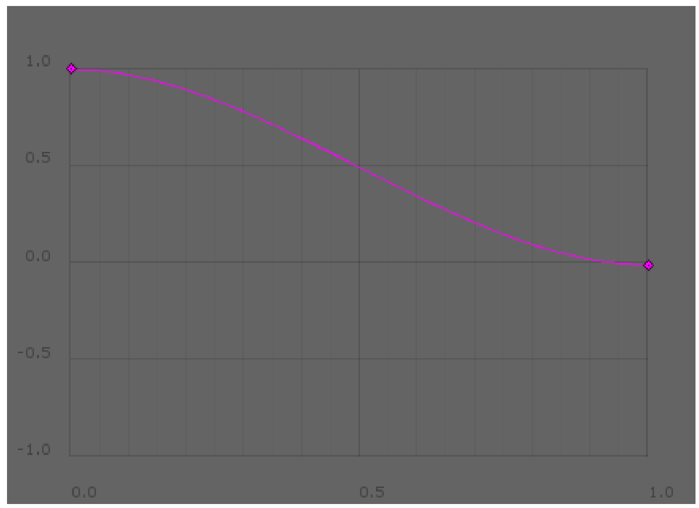
	Mode	
Axis	`PointToEdge`	`DistanceFromEntry`
Y	Indicates a multiplier of the clock's time scale, from 1 to -1.	

	Mode	
X	0 is at the `center`; 1 is at the collider's edge.	1 is at entry; 0 is at a distance of `padding` or more from entry.

A good use for this curve is to dampen an area clock's effect to make it seem more natural. Think of the curve's left being the centermost part of the clock, and right being the edge. Common damping curves for both modes are illustrated below.



A common damping curve for the PointToEdge mode.



A common damping curve for the DistanceFromEntry mode.

```
Vector center { get; set; }
```

The center of the area clock. This value is only used for the `PointToEdge` mode.

```
float padding { get; set; }
```

The padding of the area clock. This value is only used for the `DistanceFromEntry` mode.

Methods

```
void Release(Timeline timeline)
```

Releases the specified timeline from the clock's effects.

```
void ReleaseAll()
```

Releases all timelines within the clock.

> If the timeline enters the clock after having been released, it will be captured again. To configure which area clocks capture which timelines, use Unity's collision matrix .

```
void CacheComponents()
```

The components used by the area clock are cached for performance optimization. If you add or remove the `Collider` on the GameObject, you need to call this method to update the area clock accordingly.

Notes

This section contains diagrams that help understand how the `PointToEdge` and `DistanceFromEntry` modes function. For simplicity, 2D area clocks are represented, however the same concepts and calculations apply to 3D area clocks.

Point To Edge

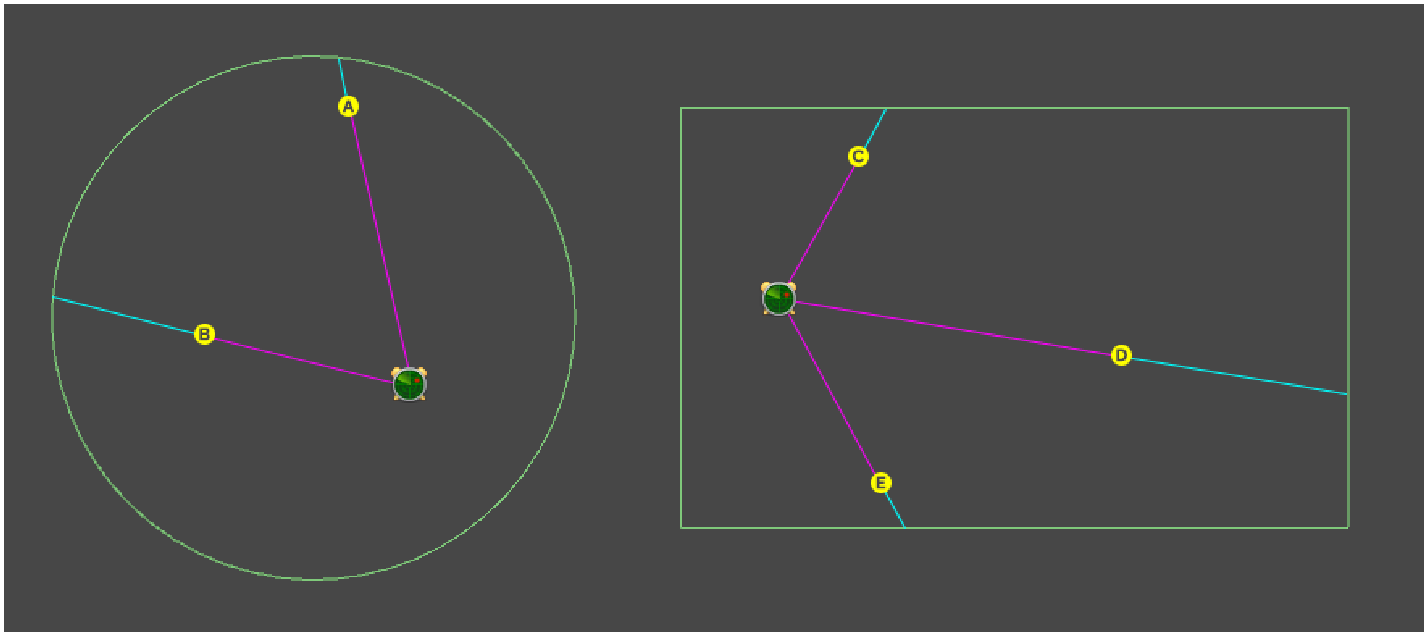


Diagram for objects within a PointToEdge area clock. Area clock icon: Clock's center Green edge: Clock's collider Yellow dot: Object's position Magenta line: Distance between center and object Cyan line: Distance between object and edge

In the point to edge mode, the X value of the `curve` property is calculated by the distance of the object from `center` to the edge of the area clock.

In the above diagram, this represents the following ratio:

x = Magenta line / (Cyan line + Magenta line)

For example, A and E would have an x value of about 0.8, while B and D would have an x value of about 0.6. In the unlikely event that an object was placed at the exact center of the clock, its x value would be 0.

Distance From Entry

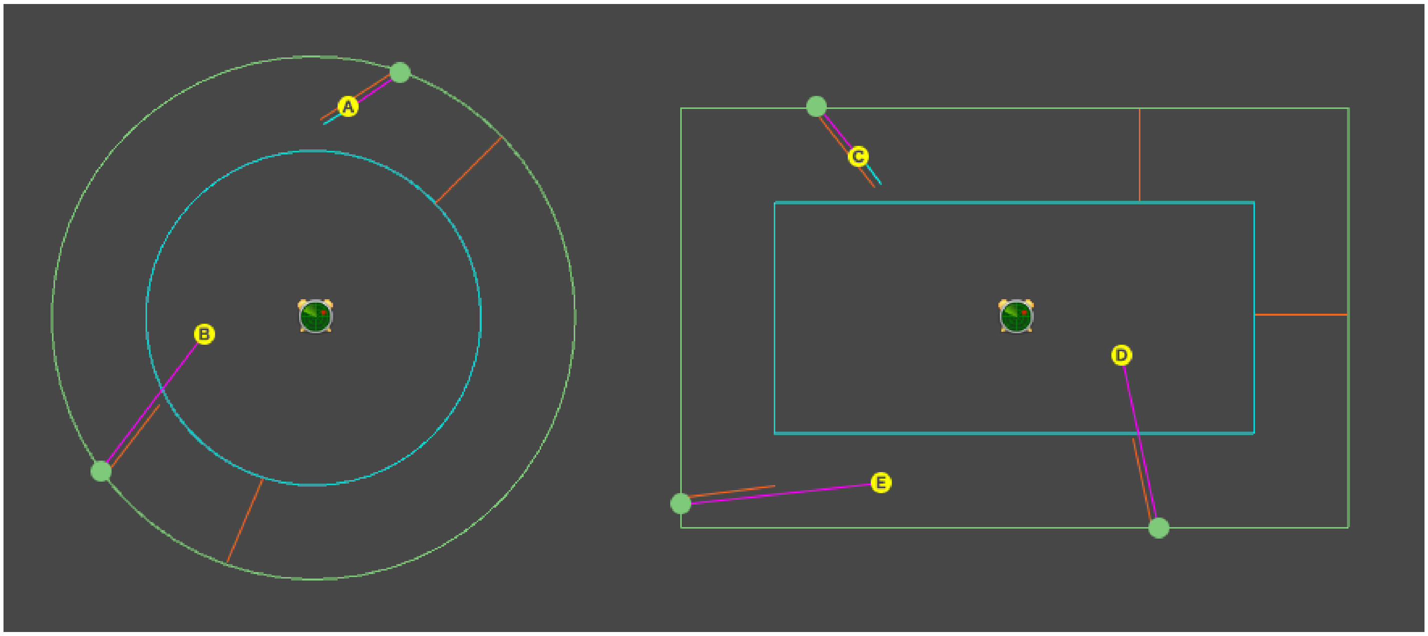


Diagram for objects within a DistanceFromEntry area clock. Area clock icon: Clock's local position Green edge: Clock's collider Green dot: Object's entry point Yellow dot: Object's position Orange line: Length of padding Cyan edge: Collider inset by padding (for reference only) Magenta line: Distance between entry and object Cyan line: Distance remaining before end of padding

In the distance from entry mode, the X value of the `curve` property is calculated by the distance of the object from its entry point relative to the value of `padding`.

In the above diagram, this represents the following ratio:

```
x = Cyan line / Orange line
```

For example, A and C would have an x value of about 0.25, while B, D and E would have an x value of 0, because they went over the maximum distance.

⚠️ The inset cyan collider has no real impact on the calculation. It is merely a helper that lets you visualize how big the value of `padding` looks while in the editor. Indeed, unless objects enter perfectly ortogonally to the collider's edge, their max distance will not be at the edge of the inset cyan collider.

✅ You can display the cyan and magenta lines for both modes in scene view by enabling `Timekeeper.debug`.

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